NLP vs. LLM for News Sentiment Analysis of Corporate Bonds

Duke University

Athena Ru

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Contents

Abstract

This study investigates whether news sentiment - quantified using both traditional natural language processing (NLP) techniques and large language models (LLMs) - can serve as a useful predictor of yield to maturity (YTM) for corporate bonds. A secondary analysis explores whether specific bond attributes influence the sentiment-YTM relationship. Historical bond data and characteristics were obtained from Refinitiv Analytics while news headlines were obtained from ProQuest TDM Studio. The bond sample consists of 135 plain vanilla, fixed-coupon debt issued in US dollars by 14 companies, divided into two groups: the Magnificent 7 (major tech companies including Tesla, Nvidia, Microsoft, Meta, Alphabet, Amazon, and Apple) and the Other 7 (diverse large-cap companies including Exxon Mobil, Walmart, Visa, Netflix, Eli Lilly, Berkshire Hathaway, and Broadcom). Vector Autoregression (VAR) modeling revealed that, for a select few bonds, news sentiment is indeed statistically significant associated with change in YTM. Secondary analysis with linear mixed models (LMMs) suggests that bond-specific attributes were not associated with the sentiment-YTM relationship. Overall, while sentiment analysis is useful for forecasting YTM in specific cases, further research including alternative models and additional predictors is needed for broader applicability.

1 Introduction

We aim to investigate whether news sentiment as quantified by traditional NLP techniques or LLMs is associated with yield to maturity (YTM) and whether sentiment may be perceived as a “useful" predictor of YTM based on Granger causality.

1.1 Corporate Bonds

Corporate bonds are publicly traded bonds issued by corporations as a way to raise debt financing (SEC Office of Investor Education & Advocacy, n.d.). Corporate bonds are taxable bonds that are considered medium- to high-quality. Investors pay an up-front lump-sum (face value) on the expectation that they will receive interest payments (coupons) over time, as well as the face value of the loan on the maturity date (the final repayment date) (Berk et al., 2012); the return on investment is primarily generated through interest payments. Figure 1.1 shows a timeline of cash flows for a 5-year corporate bond with fixed coupons.

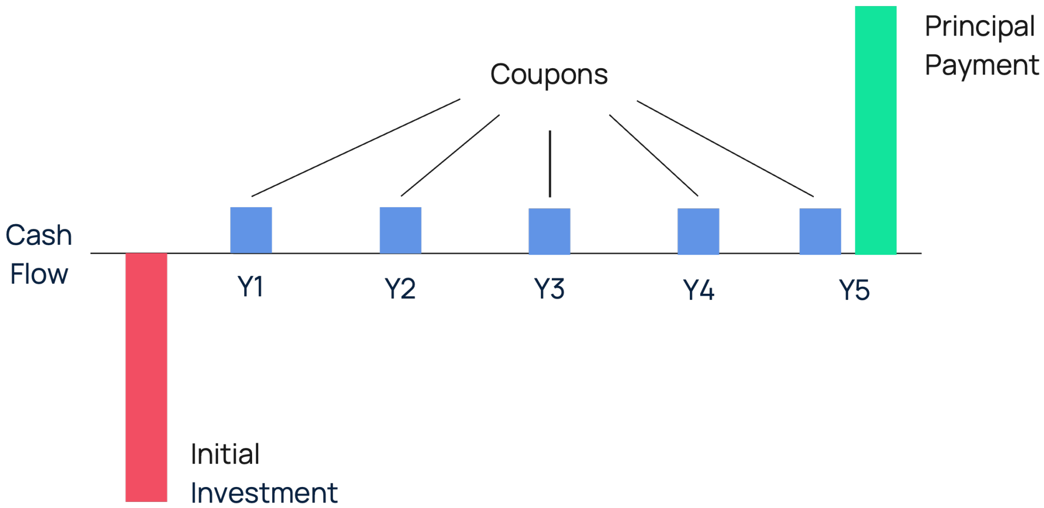

Figure 1.1: Corporate Bond Cash Flow For a Lender

For example, a \$1000 bond with a 10% coupon rate and annual payments over 5 years will pay coupon payments of \$100 every year. If this bond was bought at \$1000 - at par - and the price stayed at \$1000 over the entire life of the bond, the (annual) yield would be $\$100/\$1000 = 10\%$. Clearly, when market interest rates go up, so do bond yields. However, the value of cash flow from the bond would decrease due to inflation so investors demand a lower price. This means bond prices are always inversely related to bond yields (Berk et al., 2012). Note that if the yield on a fixed-coupon bond goes up, the borrower does not have to pay more interest. The price would go down but the payments are fixed - a new investor would simply receive a higher yield. Importantly, a higher yield is not necessarily more desirable as it simply indicates there is greater risk associated.

There are many types of bonds. Some have embedded options that allow the investor to demand early repayment ("put") or the issuer to redeem the bond before the maturity date ("call") (Financial Industry Regulatory Authority (FINRA), n.d.). Some bonds also have a floating interest rate meaning coupon payments will not be a constant amount. These characteristics all affect the yield of the bond. This study only examines plain vanilla, fixed-coupon bonds which as the name suggests, is the most basic type of bond: fixed coupon payments, no options, and no floating rates. Due to their simple structure, Rossmiller, 2018 explains that the best yield measure for these bonds is yield to maturity (YTM). While these bonds may seem boring, they are vital for investors to hedge their riskier investments against.

The yield to maturity (YTM) of a bond is the discount rate that sets the present value of the promised bond payments equal to the current market price for the bond (Berk et al., 2012). In plain English, it is the total return an investor anticipates if they hold the bond until maturity and receive all the remaining coupon payments and principal. The following Table 1.1 is taken from the SEC Office of Investor Education & Advocacy, n.d. to illustrate how YTM changes in response to changes in price:

	Bond A	Bond B	Bond C
Price (as % of face)	90%	100%	110%
Trades at	discount	par	premium
Maturity	10yr	10yr	10yr
Face value	$1,000	$1,000	$1,000
Coupon rate	4%	4%	4%
YTM	5.31%	4.00%	2.84%

Table 1.1: How YTM Changes

As previously stated, an investor's primary source of return is coupon payments. The second source of return depends on the price the bond was purchased. If the bond was purchased at par, the only return investors will earn is from the coupons. If the bond was purchased at a premium (a price greater than the face value), investors will receive a par amount less than the initial investment. Lastly, if the bond was purchased at a discount (a price less than the face value), investors will receive a par amount higher than the initial investment. Most issuers of bonds choose a coupon rate so that the bonds will initially trade at, or very close to, par.

There are 2 factors that influence plain vanilla bond prices and yields: credit (default) risk and interest rate risk (SEC Office of Investor Education & Advocacy, n.d.). The latter is due to the macroeconomic environment and cannot be controlled. The former is determined by the issuer. A company that fails to make payments or interest payments on time is said to default. Credit risk is the risk of a corporation defaulting on debt payments (Berk et al., 2012) and can be quantified by "spread" to Treasuries, which is the additional yield relative to the yield of the same maturity Treasury bond (Figure 1.2).

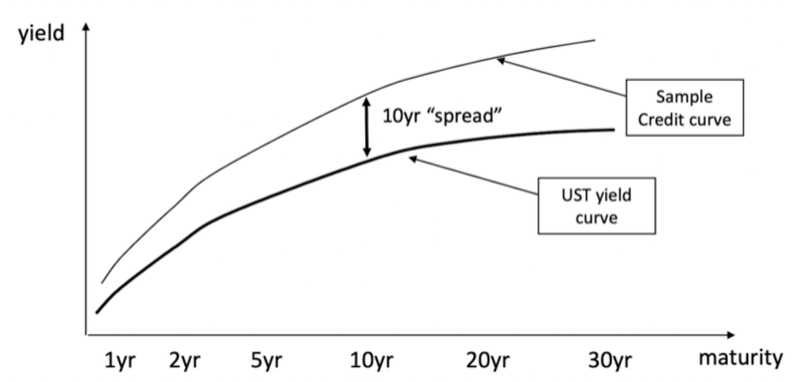

Figure 1.2: Spread of a Bond

When looking at the yield curve, the spread between Treasury curve and credit curve is greater as maturity increases because the long-term is riskier (Berk et al., 2012). Credit curves tend to "tighten" (get closer to the Treasury yield curve) in strong market environments and "widen" in recessions (higher risk of default). Spreads are largely determined by the credit rating assigned by a rating agency. The lower the credit rating, the more likely it is for a corporation to miss a coupon payment (Corporate Finance Institute (CFI), n.d.). Figure 1.3 shows the credit rating system:

	S&P ratings[1]	Moodys ratings[2]
Investment Grade	AAA	Aaa
	AA	Aa
	A	A
	BBB	Baa
High Yield (or "junk")	BB	Ba
	B	B
	CCC	Caa
	CC	Ca
	C	C

Figure 1.3: Credit Rating Systems

Insurance companies, pension funds, and some other firms/investment vehicles may only purchase Investment Grade debt. If a corporation's debt falls below BBB, investors sell, so the corporate bond price falls. Investment vehicles that can only own Investment Grade must also sell the debt, pushing the price down even further. The result is that there is a significant yield "jump" between BBB and BB. This study is only concerned with investment grade bonds.

Market perception of the credit-worthiness of an issuer directly affects prices. If investors believe an issuer does not have enough cash flow to meet payments and/or the issuer's reputation takes a hit, the price of a bond will drop and the yield will rise. Conversely, if the issuer is viewed favorably, there will be more buyers than sellers so the price will increase and the yield will drop.

1.2 Prior Work

Much research has already been conducted on equity markets. Dow Jones provides a primer for sentiment analysis of financial news, noting that negative sentiments are more related to market movements than positive ones while stronger effects are observed for mid- and small-capitalization stocks (Rodda & Nelson, 2022).

Corporate bonds likely share some of these observed effects, but there is reason to believe the effects may differ in magnitude due to the nature of corporate bonds. First, bond holders receive cash flows in the form of "coupons" (interest payments) and a final "principal" payment when the bond matures. Unlike equities, coupon and principal payments must be paid on time. If a company cannot meet its debt payments and goes bankrupt, bond holders get paid before equity holders. If a firm misses a payment, lenders can sue the company. Importantly, bond holders do not care what happens to company's share price; they only care whether firm has enough cash flow to make the interest and principal payments on its debt. This means corporate bond prices are primarily affected by the economic environment (macro factors) – especially interest rates – and concerns about the cash flow stability of the firm. Financial news articles may capture investor sentiment on the latter topic, potentially serving as a valuable predictor.

Naturally, understanding the impact of financial news sentiment on corporate bond YTM is crucial in gauging market sentiment and assessing risk. Positive news may lead to increased investor confidence, potentially driving down YTM, while negative news could signal heightened risk perceptions, leading to higher YTM. Moreover, due to the standardized nature of credit ratings, we expect less strategic manipulation of language compared to stocks.

Traditional natural language processing (NLP) methods rely on basic methods like bag-of-words and sentiment lexicons (Silge & Robinson, 2017). They need extensive labeled datasets and feature engineering for good performance. On the other hand, large language models (LLMs) break down text into tokens and output the most likely combination of tokens (IBM, 2023). This enables them to process the context of words within sentences to a greater degree than NLPs. However, because LLMs simply piece together tokens that have the highest probability of appearing next to each other, they "hallucinate" information that sounds plausible but in reality may be irrelevant, false, and/or nonsensical.

Probability & Partners, a risk advisory firm, investigated how company-specific news affect bonds issued by that company. They found that portfolios of corporate bonds whose media sentiment is improving, outperform the benchmark, even for monthly rebalancing (Borovkova & Zhang, 2022). Avoiding those bonds with declining sentiment also improves bond portfolio performance. Similarly, Cai et al., 2023 investigated relationships between corporate bond spreads for seven companies from the Euro Stoxx 50 index and macroeconomic and firm-specific news sentiment. They found that negative government and firm-specific news sentiment, in general, affect corporate bond yield spreads more than positive government and company news sentiment (Cai et al., 2023).

The effect of negative sentiment is (unsurprisingly) consistent. While we investigate similar questions as Borovkova and Zhang, 2022, the key differentiator lies in methodology. Borovkova and Zhang, 2022 use news sentiment data provided by the Refinitiv News Analytics, which employs NLP algorithms that read and interpret real-time news items. However, in addition to leveraging NLP, we will be comparing the results with large language models (LLMs) – no study has done this yet.

Many studies on corporate bonds use yield spread as the response variable. This means they are modeling credit risk. Then why not use yield spread or credit rating instead of sentiment to capture credit risk? The main problem is that credit ratings and spreads are backward-looking as they reflect historical financial performance and default probabilities. In reality, negative news about a company's debt levels may cause investors to price in higher risk before rating agencies downgrade the bonds and the spread widens. However, sentiment analysis provides a real-time, forward-looking signal of how investors perceive creditworthiness. For this reason, this study is more interested in using sentiment analysis to capture credit risk while conditioning for macroeconomic interest rate risk.

2 Methodology

2.1 Overview

We use vector autoregression (VAR) modeling to evaluate associations between the response variable of YTM and relevant predictors, including sentiment and personal consumption expenditures (PCE). Sentiment will be evaluated using two approaches: a financial sentiment lexicon for NLP-based sentiment analysis and ChatGPT for LLM-based sentiment classification. PCE, the Federal Reserve's preferred measure of inflation, is included to account for macroeconomic conditions. Together, these predictors aim to capture both credit risk and interest rate risk, key factors known to influence bond yields.

Additionally, we will conduct a secondary analysis using the coefficient estimate for sentiment from the primary model as the response variable in a linear mixed model (LMM) examining associations with bond characteristics, including type of sentiment, face value issued, lifespan, and company. This secondary analysis is aimed at exploring whether specific bond attributes are associated with any identified relationships between sentiment and YTM.

All analyses were conducted in R version 4.4.1 at a pre-defined statistical significance level of 0.05.

2.2 Data Sources & Cleaning

2.2.1 Historical Bond Data

Historical bond data and bond characteristics were obtained from Refinitiv Analytics and news headlines were retrieved from ProQuest TDM Studio. For this study, 14 companies were chosen and divided into 2 groups. The first group consists of the Magnificent 7 which includes 7 major tech companies: Tesla (TSLA), Nvidia (NVDA), Microsoft (MSFT), Meta (META), Alphabet (GOOGL), Amazon (AMZN), and Apple (AAPL). The second group consists of 7 additional companies (referred to as "Other 7") with large market capitalizations across diverse industries: Exxon Mobil (XOM), Walmart (WMT), Visa (V), Netflix (NFLX), Eli Lilly (LLY), Berkshire Hathaway (BRK), and Broadcom (AVGO). To ensure bond ratings do not introduce confounding effects, only investment grade (BBB to AAA) corporate bonds were considered.

The bond universe consists of plain vanilla, fixed-coupon debt issued in US dollars by these 14 companies and their subsidiaries. The Magnificent 7 had a total of 67 bonds that met this criteria and the Other 7 had a total of 68 bonds. Refinitiv Analytics Workspace, a leading provider of financial data (Refinitiv Analytics Team, n.d.), was used to obtain weekly historical pricing data for each bond (Table 2.1) as well as bond-specific characteristics (Table 2.2). Weekly refers to the value at the end of the week (Friday). Bonds were categorized by maturity: those with a lifespan of three years or less were classified as "Short", those with maturities greater than three but no more than ten years as "Medium", and those greater than ten years as "Long" (Investor.gov, n.d.). Lastly, personal consumption expenditures (PCE) data which is only released monthly was also pulled from Refinitiv to serve as macroeconomic data.

Variable	Definition
Yield To Maturity	Total return an investor anticipates if they hold the bond until maturity (%)
Dirty Price	Price including accrued interest
Clean Price	Price excluding accrued interest

Table 2.1: Weekly Historical Data For Each Bond

Variable	Definition
ISIN	International securities identification number
RIC	Refinitiv identification code
Parent Ticker	Ticker of parent company
Parent Issuer Name	Name of parent company
Issuer Name	Name of issuer company
Issue Date	Date bond was issued
Maturity Date	Date bond matures
Face Issued Total	Principal amount issued
Moody's Rating	Moody's announced rating
Lifespan	Short (≤ 3), Medium ($> 3, \leq 10$), Long (> 10)

Table 2.2: Bond-Level Data

Figure 2.1 shows the YTM distribution for both company groups. Notably, there are no YTM values under 0% among the Other 7 bonds. The mean YTM for both Magnificent 7 and Other 7 is 3.89%. The standard deviation of YTM for Magnificent 7 is 5.36% which is greater than that of Other 7 (1.63%). A negative YTM means the investor received less money at the maturity date than the original purchase price of the bond even after all coupon payments. Negative YTMs are infrequent but can occur when there is particularly high demand for a bond (driving the price up) and/or when interest rates are very low. This means the investor lost money. Figure 2.2 shows a zoomed-in density plot of the distribution of weekly YTM between 0% and 15%. Most YTM values are between 1% and 6% among the Other 7 while most YTM values are between 1% and 6% and 7% and 9% among the Magnificent 7.

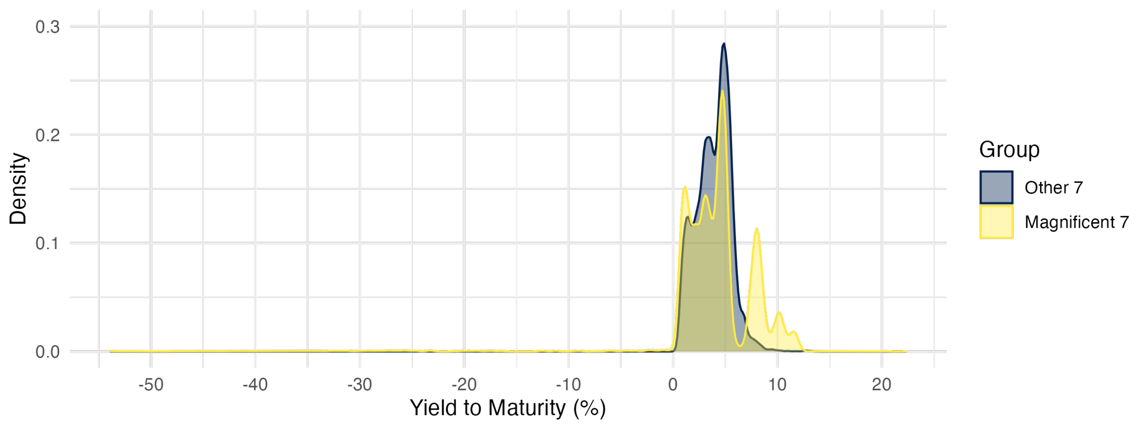

Figure 2.1: Density of Weekly YTM by Company Group

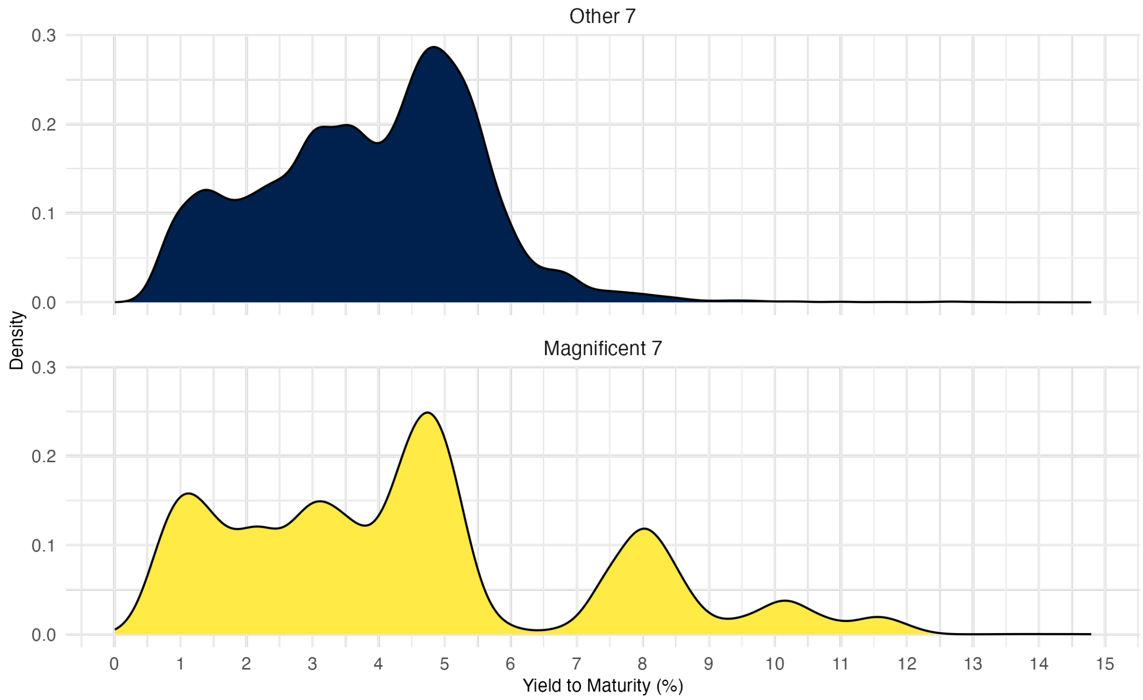

Figure 2.2: Density of Weekly YTM by Company Group (Zoomed-In)

Figures 2.3 and 2.4 show the weekly YTM time series for each bond in the two company groups. Most bonds within each company tend to follow similar trends, though some bonds experienced spikes or dips. Figures 2.5 and 2.6 show that the only bond that experienced a permanent deviation was a bond issued by Nuance Communications, a software company acquired by Microsoft in March 2022. Unfortunately, the company is now private, and no further information on why YTM plummeted for this bond is available.

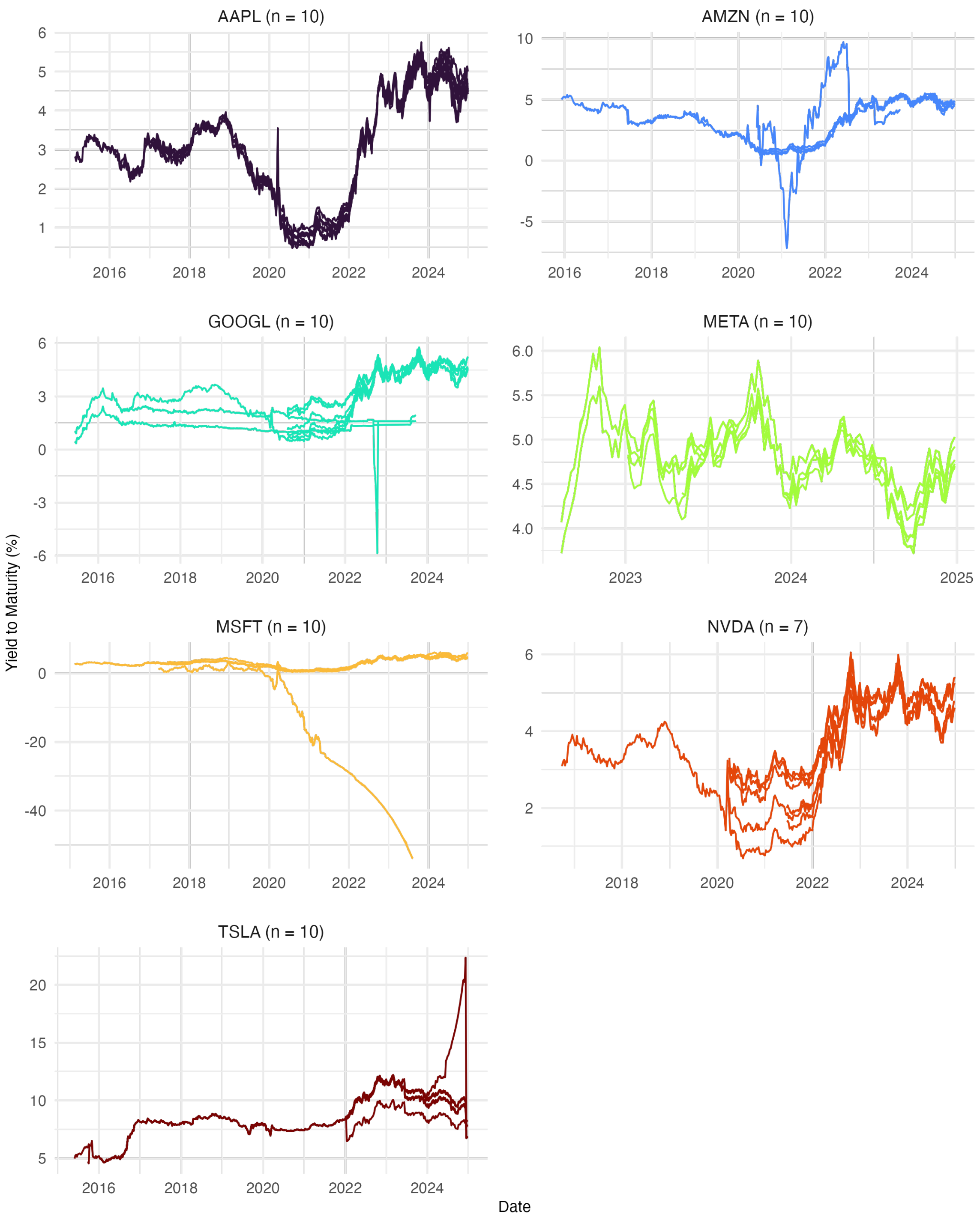

Figure 2.3: Time Series of Weekly YTM for Each Mag 7 Bond

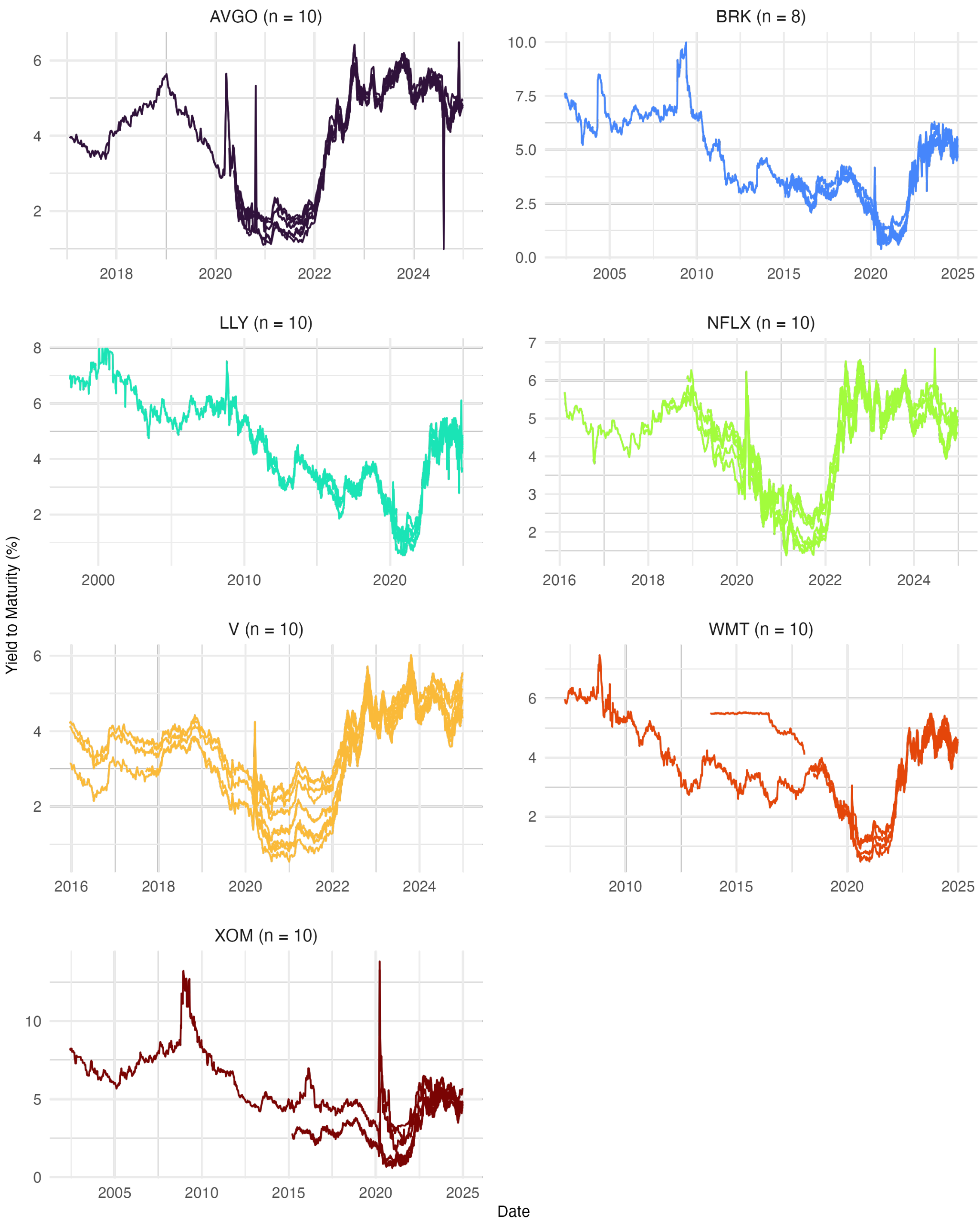

Figure 2.4: Time Series of Weekly YTM for Each Other 7 Bond

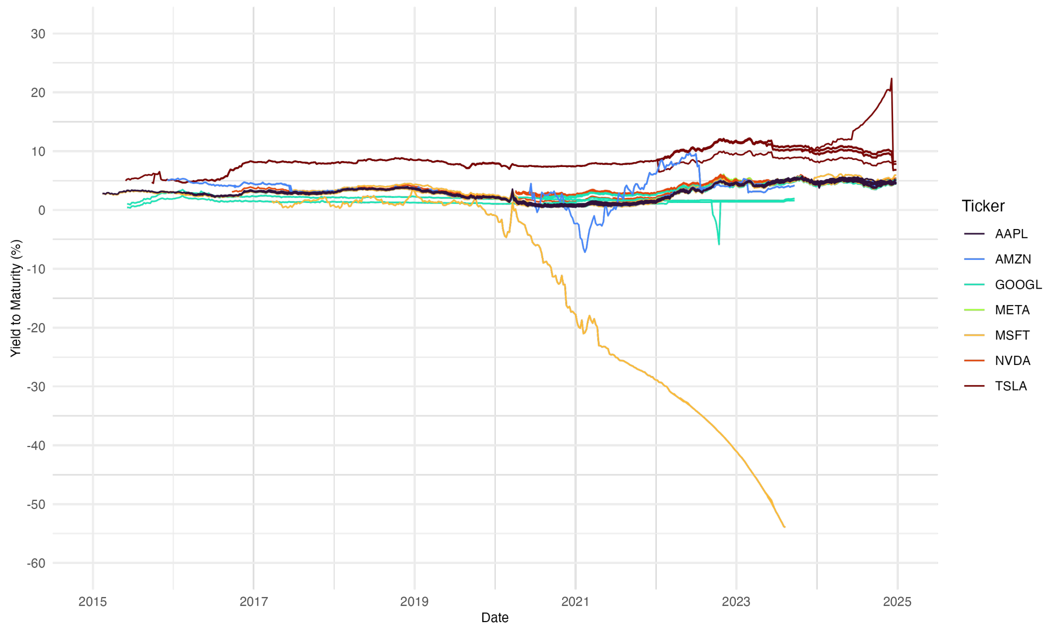

Figure 2.5: Time Series of Weekly YTM for All Mag 7 Bonds

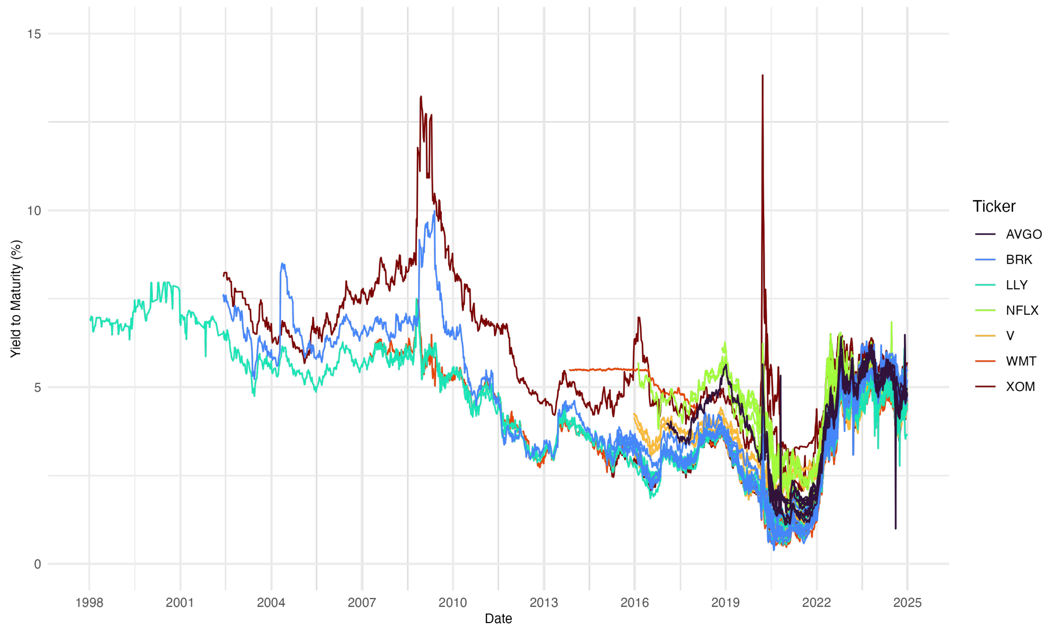

Figure 2.6: Time Series of Weekly YTM for All Other 7 Bonds

2.2.2 News Headlines

News headlines from the earliest bond issuance date through Dec 31, 2024 were pulled from ProQuest TDM Studio. The search query included headlines containing the name of the parent issuer but excluded those with keywords relating to stock performance (see Appendix A for full search query). Duplicate headlines such as updates or indicators of continued articles were removed. Encoding artifacts such as “amp;” were corrected to display the proper ampersand symbol.

To filter out irrelevant headlines (e.g., those referring to the fruit apple rather than the company), only entries where the company name appeared in the designated company column were kept. Rows with Facebook were classified under Meta and rows with Google were classified under Alphabet to account for company re-branding. Table 2.3 shows the number of cleaned news headlines for each company. Notably, the Magnificent 7 companies have a significantly higher volume of headlines, which aligns with their prominence in media coverage.

Mag 7	**Headlines Count**	**Other 7**	**Headlines Count**
Apple	10348	Walmart	6900
Amazon	9810	Netflix	6025
Tesla	6798	Eli Lilly	3372
Microsoft	6495	Berkshire Hathaway	2418
Nvidia	4577	Broadcom	1815
Alphabet	4531	Visa	1124
Meta	1337	Exxon	864

Table 2.3: Number of News Headlines For Each Company

2.3 LLM Sentiment

ChatGPT 4o-mini was used to classify news headlines on a sentiment scale of 1 to 5 (1=Strongly Negative; 2=Moderately Negative; 3=Neutral; 4=Moderately Positive; 5=Strongly Positive). See Appendix B for the prompt used. To stay within token limits, the API was called in batches of 150 headlines. ChatGPT-4o-mini was chosen over ChatGPT-4o to improve efficiency, as classification is a relatively quick task, and using 4o would have significantly increased processing time.

For unknown reasons, ChatGPT occasionally skipped headlines. Despite efforts to recover as many as possible, a small portion (<5%) of the original headlines was lost. These missing headlines are assumed to be completely missing at random and are not expected to bias subsequent analysis. The final news dataframe for each company consisted of the date, headline, and sentiment score. To align with weekly YTM data, weekly LLM sentiment for each company was computed by summing up the total sentiment over a week and dividing by the number of articles released that week.

Figure 2.7 shows the distribution of LLM sentiment for the two company groups. Most values are clustered around 3, indicating a predominantly neutral sentiment. However, the spikes at 4 are taller than those at 2, suggesting that moderately positive sentiment is more common than moderately negative sentiment.

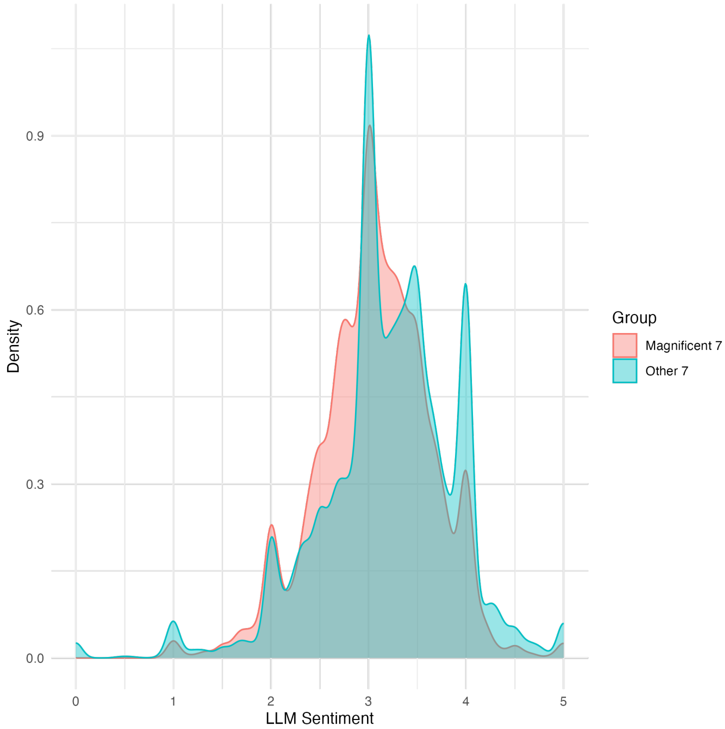

Figure 2.7: Density Plot of Weekly LLM Sentiment for Mag 7 vs Other 7 Bonds

2.4 NLP Sentiment

The Loughran-McDonald sentiment lexicon was used, as it is specifically designed for financial documents (Loughran & McDonald, 2011). Each headline was assigned a numerical positive and negative score, with the overall sentiment calculated as the difference between the two (positive score - negative score). Similarly, the weekly sentiment for each company was calculated by summing up the total sentiment over a week and dividing by the number of articles released that week.

Figure 2.8 shows the distribution of NLP sentiment for the two company groups. Most values are clustered at or around 0, which is not ideal for modeling. However, the overall trends align closely with the sentiment distribution observed in the LLM-based analysis.

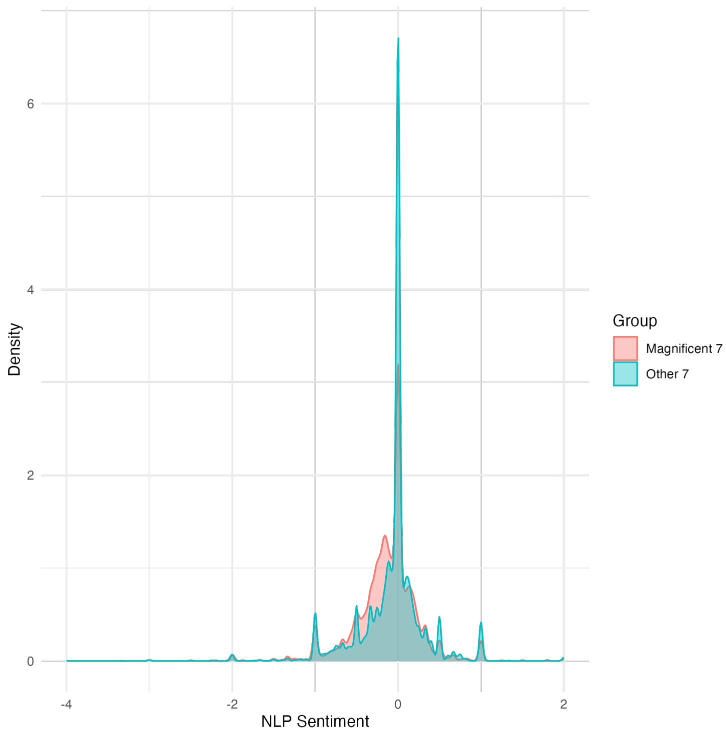

Figure 2.8: Density Plot of NLP Sentiment for Mag 7 vs Other 7 Bonds

2.5 VAR Modeling of YTM

The weekly sentiment scores were merged with the weekly YTM data by company and date. Some weeks had missing sentiment scores due to a lack of relevant news articles. To address this, a Kalman filter, which is well-suited for imputing missing values in time-series data (Moritz & Bartz-Beielstein, 2017), was applied. The final modeling dataset for each company is summarized in Table 2.4.

Variable	**Definition**
Date	Date (Friday of every week)
YTM	Weekly yield to maturity (YTM)
Bond RIC	RIC of bond
NLP Sentiment	Weekly NLP sentiment score
LLM Sentiment	Weekly LLM sentiment score
Face Issued (Bil)	Principal amount issued in billions
Lifespan	Short (≤ 3), Medium ($> 3, \leq 10$), Long (> 10)
PCE	Monthly PCE value

Table 2.4: Modeling Data for Each Company

A time series object was created for each company's modeling data using `xts` from the `xts` package. The Augmented Dickey-Fuller (ADF) test was conducted on each bond time series to check for stationarity. If the p-value was at least 0.05, first-order differencing was applied to the corresponding variable.

Vector Autoregression (VAR) modeling was selected, as both the response variable (YTM) and predictors (sentiment and PCE) were time series data that could influence each other (Pfaff, 2008). VAR models assume every variable influences every other variable in the system. This means there is an equation for each variable in the system. While lacking a theoretical structure determined by economic theory, VARs are useful for forecasting variables, testing whether one variable is useful in forecasting another (Granger causality), impulse response analysis, and forecast error variance decomposition (Hyndman & Athanasopoulos, 2021). For each bond, two VAR models with lag p were estimated using the `VAR` function from the `vars` package. The model equations with YTM as the response are shown below:

1. $YTM_t = \alpha_1 + \sum_{i=1}^{p} \beta_i YTM_{t-i} + \sum_{i=1}^{p} \gamma_i llm\_sent_{t-i} + \sum_{i=1}^{p} \delta_i pce_{t-i} + \epsilon_t$
2. $YTM_t = \alpha_2 + \sum_{i=1}^{p} \beta_i YTM_{t-i} + \sum_{i=1}^{p} \gamma_i nlp\_sent_{t-i} + \sum_{i=1}^{p} \delta_i pce_{t-i} + \epsilon_t$

`VARselect` was used to select the optimal lag for each model based on the Akaike Information Criterion (AIC). As the model is already sparse with only 2 predictors, AIC was used. The coefficient estimate and p-value for the first lag of the sentiment variable were extracted from the model with YTM as the response.

To assess model fit, diagnostic tests were performed, including the serial correlation test, ARCH test, normality tests, and stability checks. `serial.test` was used to conduct the serial correlation test. If the p-value is greater than 0.05, then there is no serial correlation in the VAR model. `arch.test` was used to conduct the ARCH test. This checks for heteroscedasticity (non-constant variance) because volatility in financial markets tends to be non-constant. If the p-value is greater than 0.05, then the model does not suffer from heteroscedasticity. `normality.test` was used to check if the residuals are normally distributed. If the p-value is greater than 0.05, then the model passes this test. If the normality test fails, it is not the biggest violation. `stability` using OLS cumulative sum as the type was used to test for structural breaks in the residuals. If after plotting the stability check, there are no points beyond upper and lower bounds, then the system is stable and there are no structural breaks in the residuals which is desirable.

`causality` was used to conduct granger causality on the models that passed the serial correlation test, ARCH test, and normality JB test. If normality JB failed but stability passed, then the model will still be included. The null hypothesis is that sentiment does not Granger-cause YTM. If the p-value is not less than 0.05, then we fail to reject the null hypothesis. If the p-value is less than 0.05, indicating that sentiment is a useful predictor of YTM, then impulse response analysis (`irf`) and forecasting (`predict`) will be conducted on the model.

2.6 Predicting VAR Sentiment Relationships

A secondary analysis was conducted using the VAR sentiment coefficient estimate as the response variable, with bond-level data as predictors. Bonds issued by the same parent company might share similar characteristics such that this company-specific relationship violates independence. Thus, a linear mixed model (lmm) was used. The `lmer` function from the `lmerTest` package was used to fit the models (Kuznetsova et al., 2017). The response variable is the sentiment coefficient estimate from the VAR model. Fixed effects consisted of the sentiment type, lifespan, and whether YTM was differenced. The random effect was face issued (in billions) grouped by company. Figure 2.9 and Figure 2.10 confirm the potential heterogeneity between different issuers.

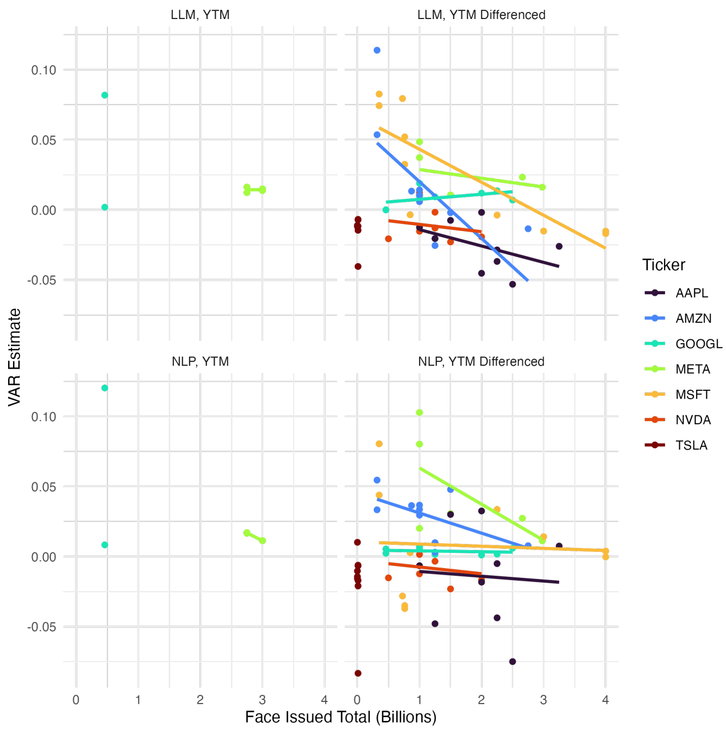

Figure 2.9: VAR Coefficient Estimate vs. Face Issued for Mag 7 Bonds

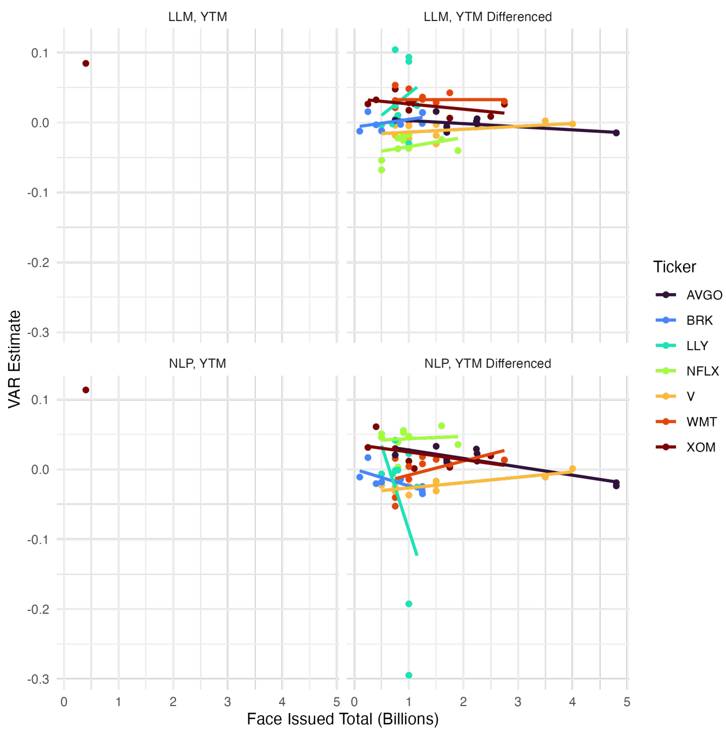

Figure 2.10: VAR Coefficient Estimate vs. Face Issued for Other 7 Bonds

For the most part, the fitted lines are in the same direction between LLM and NLP plots within a company group. Exceptions are Berkshire Hathaway, Eli Lilly, and Walmart. ANOVA tests and residual plots were used to evaluate the model fit and determine the significance of the predictors.

3 Results

3.1 VAR Results

134 VAR models were run for the Magnificent 7 and 136 were run for the Other 7. Figure 3.1 shows the distributions of VAR coefficient estimates for the sentiment variable from the first lag by company group.

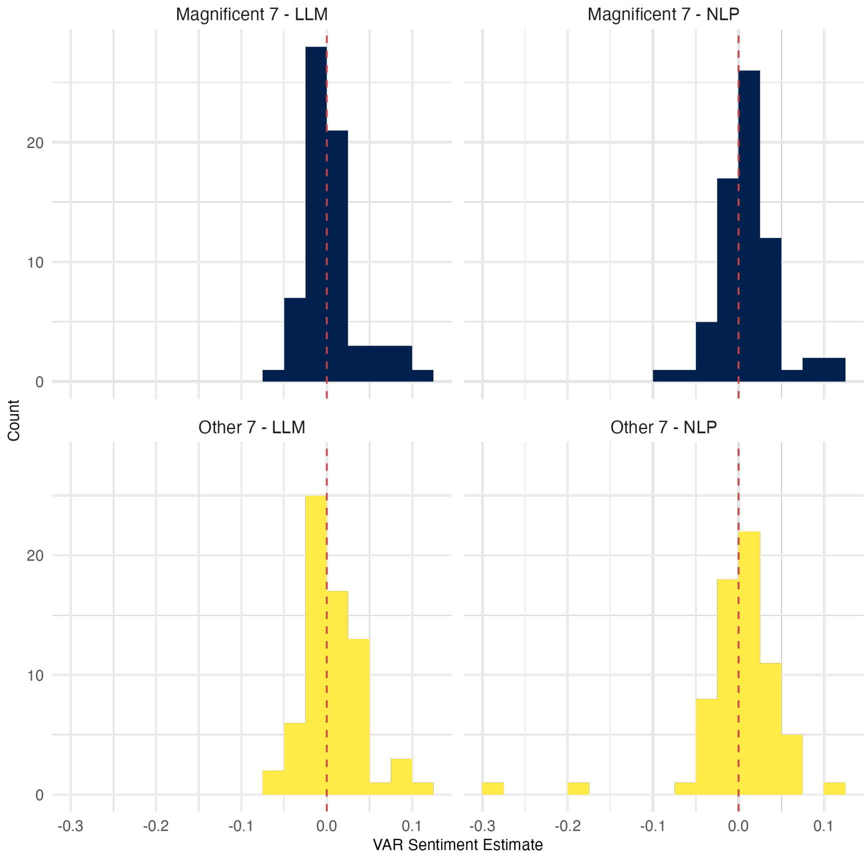

Figure 3.1: VAR Sentiment Estimates by Company Group and Sentiment Analysis

The distributions are roughly normal centered and around 0. The small magnitude of the estimates is to be expected as YTM is measured in percentages and does not generally have large fluctuations. It is important to keep in mind that some variables were differenced so a positive sentiment estimate does not necessarily mean we expect an increase in YTM.

The only sentiment estimates that were statistically significant at the $\alpha = 0.05$ level among the Magnificent 7 bonds was an Alphabet bond (ISIN = US31816QAD34) issued by Mandiant in 2015. The sentiment coefficient was statistically significant in both VAR models for this bond. Among the Other 7 bonds, 6 VAR models using LLM sentiment had a statistically significant sentiment coefficient at the $\alpha = 0.05$ level. These were 3 Exxon Mobil bonds (ISIN = US723787AT45, US30231GBJ04, US30231GBH48), 2 Walmart bonds (ISIN = US931142EM13, US931142EE96), and 1 Netflix bond (ISIN = USU74079AT84). These estimates and p-values are summarized in Table 3.1. See Table C.1 in Appendix C for full table of estimates.

ISIN	Ticker	Estimate	p-value	Sentiment Type
US31816QAD34	GOOGL	0.082	0.030	LLM
US31816QAD34	GOOGL	0.120	0.009	NLP
USU74079AT84	NFLX	-0.068	0.025	LLM
US931142EE96	WMT	0.030	0.044	LLM
US931142EM13	WMT	0.036	0.040	LLM
US30231GBH48	XOM	0.026	0.036	LLM
US30231GBJ04	XOM	0.028	0.022	LLM
US723787AT45	XOM	0.048	0.002	LLM

Table 3.1: VAR Models with a Significant Sentiment Predictor

The estimated regression equations for the Netflix bond are:

$$\widehat{diff(YTM)}_t = 0.165 - 0.184 \times diff(YTM)_{t-1} - 0.068 \times LLM\_sent_{t-1} + 0.172 \times PCE_{t-1}$$

$$\widehat{LLM\_sent}_t = 2.452 + 0.080 \times diff(YTM)_{t-1} + 0.220 \times LLM\_sent_{t-1} - 0.105 \times PCE_{t-1}$$

$$\widehat{PCE}_t = 0.040 + 0.026 \times diff(YTM)_{t-1} - 0.0002 \times LLM\_sent_{t-1} + 0.877 \times PCE_{t-1}$$

Next, we check diagnostics to determine which models to forecast.

3.1.1 Diagnostics

To assess model fit, the serial correlation test, ARCH test, normality tests, and stability checks were applied on each VAR model. Table D.1 in Appendix D shows the proportion of models that passed each test (p-value > 0.05) by company and sentiment type. Of 270 models, 84.1% passed the serial correlation test, 40.7% passed the ARCH test, 4.1% passed the JB normality test, and 52.6% passed the stability test.

Models that passed the serial correlation test, ARCH test, and normality JB test will be used to test for Granger causality. If normality JB failed but stability passed, then the model will still be included. A total of 65 models (31 LLM and 34 NLP) out of 270 met this criteria.

3.1.2 Granger Causality

Granger causality was conducted on the 65 models. 3 models had a p-value less 0.05, meaning for these models, we reject the null hypothesis that sentiment does not Granger-cause YTM. These 3 models were for a Microsoft bond (ISIN = USU59340AH90) using NLP to calculate the sentiment score, another Microsoft bond (ISIN = US594918CG78) using NLP, and a Walmart bond (ISIN = US931142FB49) using LLM to calculate the sentiment score. Sentiment is a useful predictor of YTM (differenced) for these 3 bonds.

Next, impulse response analysis was conducted on these 3 models to see what would happen to YTM (differenced) if a shock occurred to sentiment. Figure 3.2 shows that a positive shock in NLP sentiment is associated with a decrease in change in YTM over the next 2 weeks for these two Microsoft bonds. Afterwards, the change in YTM recovers and levels out. The dashed red lines indicate the 95% confidence interval and includes 0 so there could be no effect on change in YTM.

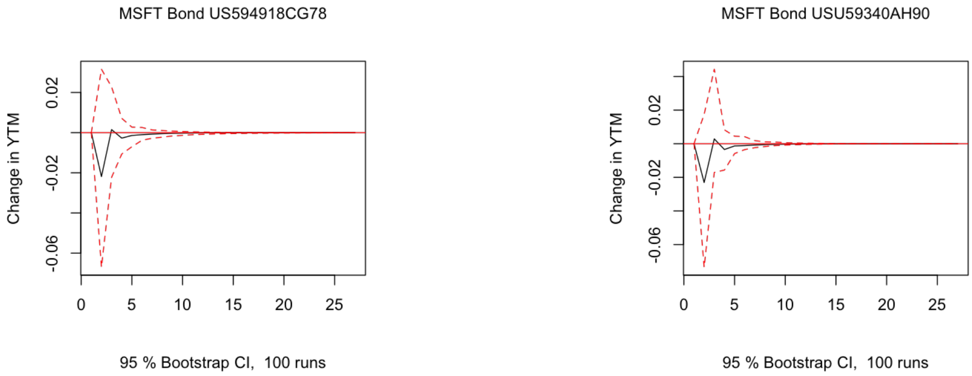

Figure 3.2: Shock from NLP Sentiment

Figure 3.3 shows that a positive shock in LLM sentiment is associated with an increase in change in YTM over the first 2 weeks followed by a sharp drop over the third week for the Walmart bond.

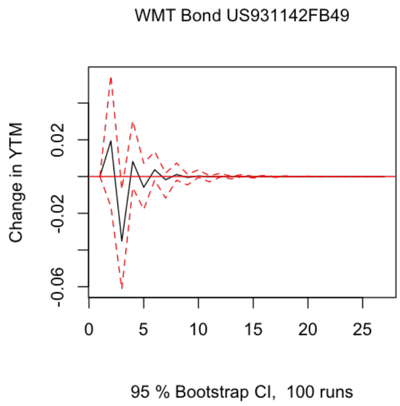

Figure 3.3: Shock from LLM Sentiment

Figure 3.4 shows the forecast 4 weeks ahead. The black gradient shows the 95% confidence interval. The y-axis ranges from -0.4 to 0.3 because differencing was applied to YMT for these bonds to ensure stationarity. For Microsoft Bond US594918CG78 and USU59340AH90, the change in YTM is expected to be around -0.05 over the next 4 weeks. For Walmart bond US931142FB49, the change in YTM is expected to decrease further before rebounding.

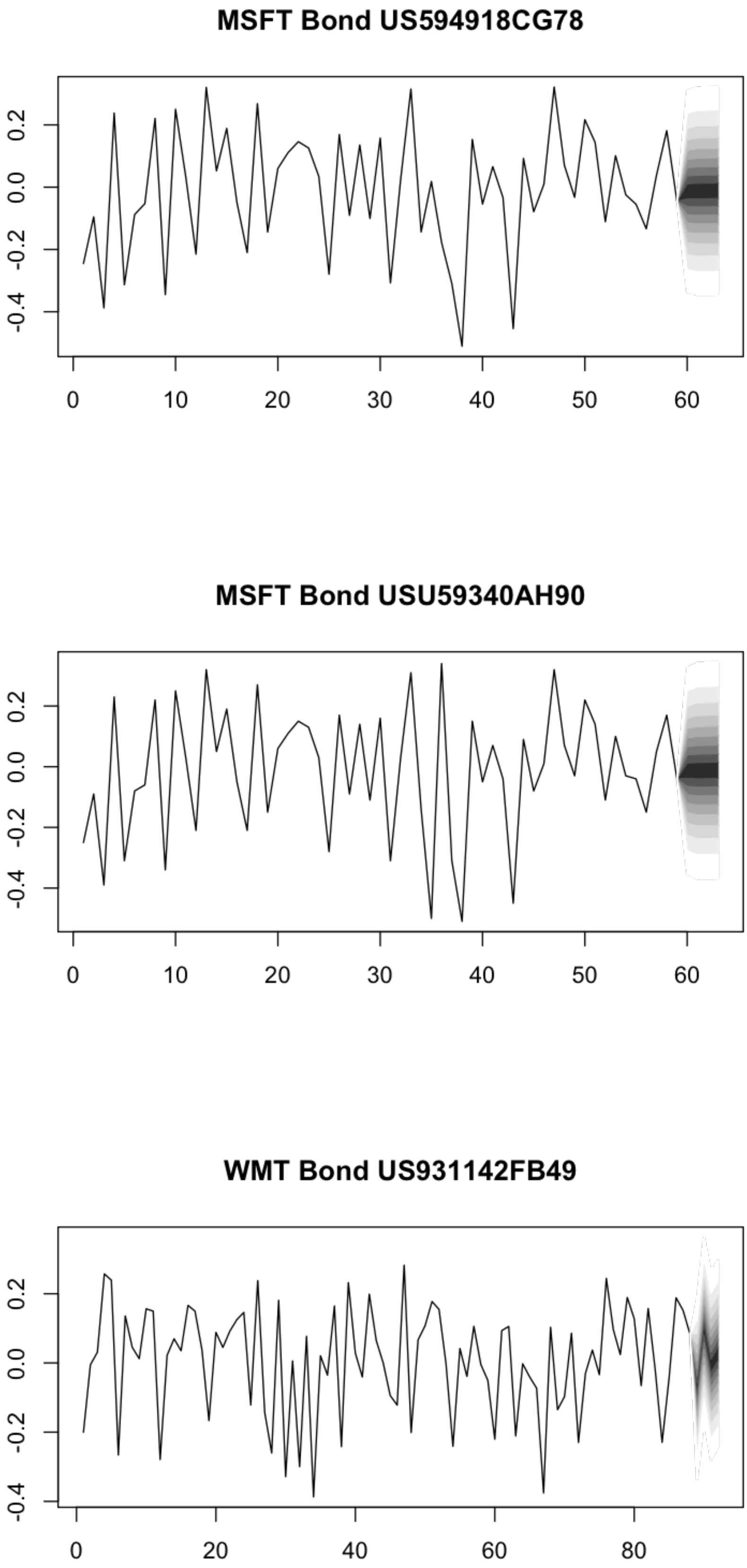

Figure 3.4: Change in YTM Forecast

Figure 3.5 shows predictions from the forecast with the truth from the first 3 weeks of 2025 (all the time-series data used for VAR modeling ended in 2024). The predictions up to the second week were most accurate for the Walmart bond that used LLM sentiment score.

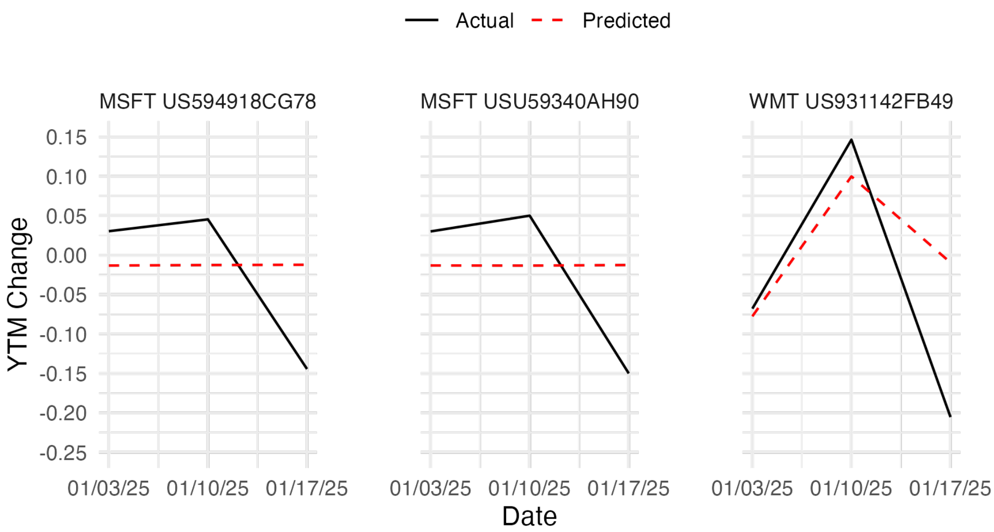

Figure 3.5: Predictions vs. Actual YTM Changes

3.2 LMM Results

The first linear mixed model was run on the Magnificent 7 sentiment coefficients. Fixed effects consisted of the sentiment type, lifespan, and whether YTM was differenced. The random effect was face issued (in billions) grouped by company. Table 3.2 shows the coefficient estimates. Among the fixed effects, only whether YTM was differenced was statistically significant at the $\alpha = 0.05$ level. ANOVA results also confirmed this. Figure 3.6 shows that the plot of residuals do not exhibit constant variance.

Ticker	**Intercept**	**Face Issued (B)**	**Sentiment: LLM**	**Lifespan: Medium**	**Lifespan: Long**	**YTM Diff**
AAPL	-0.014	0.003				
AMZN	0.048	-0.014	⋮	⋮	⋮	⋮
GOOGL	0.029	-0.009				
META	0.066	-0.020	-0.003	0.015	0.011	-0.019
MSFT	0.051	-0.015				
NVDA	-0.002	$<$ -0.001	⋮	⋮	⋮	⋮
TSLA	-0.009	0.002				

Table 3.2: LMM Fixed Effects Estimates for Mag 7

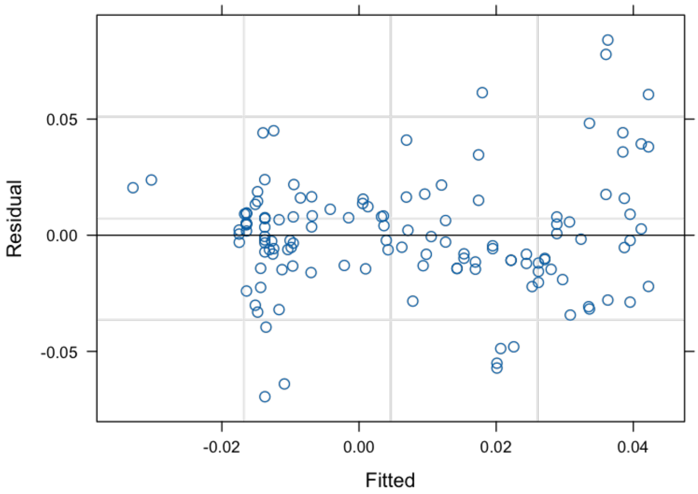

Figure 3.6: Residual Plot of Mag 7 LMM

For Meta bonds, we expect the sentiment coefficient to decrease by 0.02 for every 1 billion increase in face issued, holding all else constant. For all Magnificent 7 bonds, we expect the sentiment coefficient to decrease by 0.003 if the VAR model used LMM to calculate sentiment instead of NLP, holding all else constant.

The same model was run on the Other 7 sentiment coefficients but without random slopes and with face issued value in billions being used as a fixed effect due to estimability concerns. Table 3.3 shows the coefficient estimates. Similar to the first model, only whether YTM was differenced was statistically significant at the $\alpha = 0.05$ level, confirmed via ANOVA. Figure 3.7 suggest that the plot of residuals do not exhibit constant variance.

Ticker	**Intercept**	**Face Issued (B)**	**Sentiment: LLM**	**Lifespan: Medium**	**Lifespan: Long**	**YTM Diff**
AVGO	0.085					
BRK	0.077	⋮	⋮	⋮	⋮	⋮
LLY	0.077					
NFLX	0.084	-0.002	0.007	0.002	-0.002	-0.083
V	0.074					
WMT	-0.090	⋮	⋮	⋮	⋮	⋮
XOM	-0.094					

Table 3.3: LMM Fixed Effects Estimates for Other 7

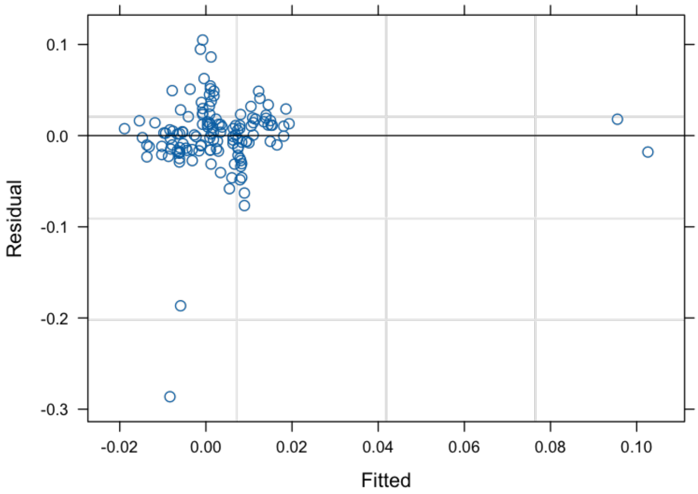

Figure 3.7: Residual Plot of Other 7 LMM

For all Other 7 bonds, we expect the sentiment coefficient to decrease by 0.002 for every 1 billion increase in face issued, holding all else constant. We also expect the sentiment coefficient to increase by 0.007 if the VAR model used LMM to calculate sentiment instead of NLP.

4 Discussion

VAR models were used to analyze on YTM, sentiment, and PCE time series data. 8 models identified sentiment as a significant predictor when YTM (differenced) was the response. Diagnostics revealed that 65 models passed enough statistical checks to be considered a useful model. However, Granger causality further showed that sentiment was a useful predictor of YTM (differenced) in only 3 of the 65 models. Impulse response analysis and forecasting showed that the predictions were moderately accurate up to the second week. Notably, for Walmart bond US931142FB49, forecasts that used LLM-derived sentiment closely matched observed changes in YTM, demonstrating the potential value of LLM sentiment in predicting YTM for this bond.

This study aimed to assess whether news sentiment as quantified by traditional NLP techniques or LLMs is a useful predictor of YTM. VAR modeling results suggest that for a select few bonds, news sentiment is indeed a useful predictor of YTM (differenced). In particular, for Walmart bond US931142FB49, LLM sentiment was a more useful predictor than NLP sentiment, as evidenced by the accurate predictions. However, for Microsoft bonds US594918CG78 and USU59340AH90, NLP sentiment was a more useful predictor than LLM sentiment. This shows that the effectiveness of sentiment analysis techniques may vary depending on the bond issuer and market conditions.

Out of 270 VAR models, only 3 were useful for conducting impulse response analysis and forecasting. This finding underscores the challenges in using sentiment as a broad predictor of YTM changes and highlights the need for further refinement in model selection, variable inclusion, and sentiment quantification techniques. While sentiment analysis can provide valuable insights for certain bonds, its predictive utility remains limited across a wider range of fixed-income securities.

LMM models were conducted to see if specific bond attributes influence the relationship between sentiment and YTM. Unfortunately, none of the bond attribute variables were found to be statistically significant predictors. This suggests the influence of sentiment on YTM operates independently of these bond characteristics, or that additional factors not captured in this analysis may play a more significant role in shaping bond yields.

4.1 Limitations

This study focused exclusively on fixed plain vanilla bonds from 14 major companies, limiting the generalizability of the findings. It is possible that sentiment analysis could be more useful for predicting YTM in other types of bonds, such as convertible bonds or floating-rate notes.

One of the major challenges in this analysis was the high amount of neutral sentiment scores, particularly from traditional NLP techniques, which may have reduced the predictive power of sentiment in the models. The Loughran-McDonald sentiment dictionary which only had 4,150 terms may have been too limited in scope and missed important keywords in financial headlines that could have influenced sentiment classification. Another limitation was the use of OpenAI's 4o-mini model for efficiency. While the 4o-mini model provided a reasonable trade-off between speed and accuracy, the full 4o model may have offered better reasoning and, consequently, more accurate sentiment classification.

We did not adjust for multiplicity concerns in our analyses, given that our objective was exploratory; future confirmatory analyses on our results should ensure that family-wise type I error is maintained. There were also some issues with residual diagnostics in the linear mixed model, specifically regarding the normality assumption. A generalized linear mixed model (GLMM) could have been a better approach to account for non-normal residual distributions.

4.2 Future Work

Future research could explore Bayesian VAR (BVAR) models, which incorporate prior distributions to improve forecasting accuracy (Kuschnig & Vashold, 2021). Expanding the analysis to other types of bonds, such as high-yield bonds or bonds from a broader set of companies (e.g., the entire S&P 500), could provide further insight into whether sentiment is more impactful in different market segments. High-yield bonds, in particular, could be more influenced by sentiment since investors in these riskier securities could be more reactive to news compared to those holding investment-grade corporate bonds. While credit risk measures such as bond spreads which offer a more historical and backward-looking perspective were not used in this study, perhaps when combined with future-looking sentiment, predictive accuracy could be further improved.

Additionally, sentiment analysis could be expanded to include sector- or industry-specific news rather than focusing solely on company-level sentiment. For example, semiconductor companies may be affected by regulatory changes impacting the entire industry, which could have a significant influence on bond yields beyond company-specific sentiment. By refining sentiment classification methods and expanding the scope of the dataset, future studies may be able to uncover more robust relationships between sentiment and bond yields.

A Appendix A

Magnificent 7 Search Query

- Date Range: Feb 1, 2015 to Dec 31, 2024
- Query: TITLE(Nvidia OR Alphabet OR Amazon OR Apple OR Meta OR Microsoft OR Tesla) NOT TITLE(stock OR share OR dow OR sell? OR surrender? OR register? OR buy? OR gift? OR overweight OR underweight OR neutral OR target OR rally OR options)
- Data Source: Wall Street Journal (Online), Bloomberg Businessweek, Forbes, Dow Jones Institutional News, MarketWatch, The Motley Fool [BLOG], The Motley Fool UK [BLOG], Kiplinger [BLOG], NASDAQ OMX's News Release Distribution Channel, Benzinga Newswires

Other 7 Search Query

- Date Range: Jan 3, 2013 to Dec 31, 2024
- Query: TITLE(Broadcom OR Berkshire OR Walmart OR Lilly OR Visa OR Exxon Mobil OR Netflix) NOT TITLE(stock OR share OR dow OR sell? OR surrender? OR register? OR buy? OR gift? OR overweight OR underweight OR neutral OR target OR rally OR options)
- Data Source: Wall Street Journal (Online), Bloomberg Businessweek, Forbes, Dow Jones Institutional News, MarketWatch, The Motley Fool [BLOG], The Motley Fool UK [BLOG], Kiplinger [BLOG], NASDAQ OMX's News Release Distribution Channel, Benzinga Newswires

B Appendix B

ChatGPT-4o-mini Prompt: Classify news headlines for sentiment analysis. My sentiment scale ranges from 1 (strong negative) to 5 (strong positive). An example headline and description are given for each rating.

1. Strongly Negative - "Bitcoin Declines Will Hurt Tesla Earnings"
   - strong negative shock
2. Moderately Negative - "Tesla Earnings Likely Clouded by Shanghai Factory Shutdown – WSJ
   - not likely to do better
   - hasn't gotten significantly worse but is still not doing well
3. Neutral - "Earnings Outlook For NVIDIA"
   - factual statement
   - a neutral question about future performance
4. Moderately Positive - "GM Earnings Are Tuesday. Things Might Be Good. Thank Tesla. – Barrons.com"
   - likely to do better
   - hasn't gotten significantly better but is still doing alright
5. Strongly Positive - "Tesla Crushes Earnings Estimates. Wall Street Is Speechless."
   - strong positive shock

Using the above scale, classify each headline in the given order with a numerical sentiment score. The returned format should be headline $\sim$ score.

C Appendix C

Table C.1: VAR First Lag Sentiment Estimates

ISIN	Ticker	Estimate	p-value	Sentiment Type
US037833AZ38	AAPL	-0.008	0.646	LLM
US037833AZ38	AAPL	0.030	0.204	NLP
US037833BG48	AAPL	-0.002	0.912	LLM
US037833BG48	AAPL	0.033	0.175	NLP
US037833BY53	AAPL	-0.026	0.154	LLM
US037833BY53	AAPL	0.008	0.770	NLP
US037833BZ29	AAPL	-0.029	0.118	LLM
US037833BZ29	AAPL	-0.005	0.851	NLP
US037833DF47	AAPL	-0.031	0.110	LLM
US037833DF47	AAPL	0.006	0.833	NLP
US037833DN70	AAPL	-0.045	0.130	LLM
US037833DN70	AAPL	-0.018	0.675	NLP
US037833DT41	AAPL	-0.037	0.109	LLM
US037833DT41	AAPL	-0.044	0.184	NLP
US037833DX52	AAPL	-0.021	0.385	LLM
US037833DX52	AAPL	-0.048	0.142	NLP
US037833EB24	AAPL	-0.053	0.053	LLM
US037833EB24	AAPL	-0.075	0.057	NLP
US037833ES58	AAPL	-0.013	0.751	LLM
US037833ES58	AAPL	-0.006	0.909	NLP
US023135BN51	AMZN	0.013	0.381	LLM
US023135BN51	AMZN	0.036	0.086	NLP
US023135BQ82	AMZN	-0.025	0.107	LLM
US023135BQ82	AMZN	0.003	0.885	NLP
US023135BX34	AMZN	-0.014	0.508	LLM
US023135BX34	AMZN	0.008	0.774	NLP
US023135CE44	AMZN	-0.002	0.933	LLM
US023135CE44	AMZN	0.048	0.112	NLP
US023135CN43	AMZN	-0.021	0.417	LLM
US023135CN43	AMZN	0.010	0.751	NLP
US68269GAA58	AMZN	0.114	0.276	LLM
US68269GAA58	AMZN	0.055	0.722	NLP
US68269GAB32	AMZN	0.054	0.595	LLM
US68269GAB32	AMZN	0.033	0.815	NLP
US966837AD89	AMZN	0.011	0.498	LLM
US966837AD89	AMZN	0.034	0.126	NLP
US966837AE62	AMZN	0.014	0.382	LLM
US966837AE62	AMZN	0.037	0.107	NLP
USU96710AA34	AMZN	0.006	0.697	LLM
USU96710AA34	AMZN	0.030	0.172	NLP
US11134LAG41	AVGO	-0.015	0.339	LLM

ISIN	Ticker	Estimate	p-value	Sentiment Type
US11134LAG41	AVGO	-0.024	0.301	NLP
US11135FAM32	AVGO	-0.014	0.384	LLM
US11135FAM32	AVGO	0.010	0.666	NLP
US11135FAN15	AVGO	-0.006	0.741	LLM
US11135FAN15	AVGO	0.016	0.516	NLP
US11135FAT84	AVGO	-0.002	0.921	LLM
US11135FAT84	AVGO	0.022	0.487	NLP
US11135FBB67	AVGO	0.001	0.948	LLM
US11135FBB67	AVGO	0.029	0.374	NLP
US928563AD71	AVGO	0.003	0.935	LLM
US928563AD71	AVGO	0.020	0.715	NLP
US928563AJ42	AVGO	0.016	0.444	LLM
US928563AJ42	AVGO	0.033	0.258	NLP
USU1108LAD11	AVGO	-0.015	0.327	LLM
USU1108LAD11	AVGO	-0.019	0.399	NLP
USU1109MAM83	AVGO	0.005	0.893	LLM
USU1109MAM83	AVGO	0.012	0.837	NLP
USU1109MAJ54	AVGO	-0.007	0.703	LLM
USU1109MAJ54	AVGO	0.013	0.607	NLP
US084659AD37	BRK	-0.003	0.791	LLM
US084659AD37	BRK	-0.021	0.337	NLP
US084659AS06	BRK	0.014	0.488	LLM
US084659AS06	BRK	-0.025	0.485	NLP
US084659AT88	BRK	-0.001	0.959	LLM
US084659AT88	BRK	-0.031	0.464	NLP
US12189LAV36	BRK	-0.012	0.282	LLM
US12189LAV36	BRK	-0.018	0.329	NLP
US549271AA22	BRK	-0.012	0.232	LLM
US549271AA22	BRK	-0.011	0.528	NLP
US695114CS55	BRK	0.016	0.231	LLM
US695114CS55	BRK	0.017	0.457	NLP
US740189AM73	BRK	-0.003	0.816	LLM
US740189AM73	BRK	-0.014	0.489	NLP
USU0740LAJ45	BRK	0.014	0.513	LLM
USU0740LAJ45	BRK	-0.035	0.335	NLP
US02079KAC18	GOOGL	0.012	0.301	LLM
US02079KAC18	GOOGL	0.001	0.938	NLP
US02079KAD90	GOOGL	0.014	0.329	LLM
US02079KAD90	GOOGL	0.002	0.914	NLP
US02079KAE73	GOOGL	0.009	0.487	LLM
US02079KAE73	GOOGL	0.002	0.919	NLP
US02079KAF49	GOOGL	0.007	0.580	LLM
US02079KAF49	GOOGL	0.006	0.684	NLP
US02079KAH05	GOOGL	0.010	0.514	LLM
US02079KAH05	GOOGL	0.005	0.743	NLP
US02079KAJ60	GOOGL	0.019	0.193	LLM
US02079KAJ60	GOOGL	0.007	0.670	NLP

ISIN	Ticker	Estimate	p-value	Sentiment Type
US31816QAA94	GOOGL	0.000	0.989	LLM
US31816QAA94	GOOGL	0.005	0.373	NLP
US31816QAB77	GOOGL	0.000	0.988	LLM
US31816QAB77	GOOGL	0.002	0.571	NLP
US31816QAC50	GOOGL	0.002	0.747	LLM
US31816QAC50	GOOGL	0.008	0.239	NLP
US31816QAD34	GOOGL	0.082	0.030	LLM
US31816QAD34	GOOGL	0.120	0.009	NLP
US532457AM04	LLY	-0.003	0.783	LLM
US532457AM04	LLY	-0.007	0.752	NLP
US532457AZ17	LLY	-0.002	0.840	LLM
US532457AZ17	LLY	-0.007	0.671	NLP
US532457BH00	LLY	0.010	0.329	LLM
US532457BH00	LLY	-0.001	0.951	NLP
US532457BP26	LLY	0.005	0.693	LLM
US532457BP26	LLY	-0.002	0.931	NLP
US532457BV93	LLY	0.024	0.099	LLM
US532457BV93	LLY	-0.025	0.307	NLP
US532457CE69	LLY	0.104	0.156	LLM
US532457CE69	LLY	-0.018	0.898	NLP
US532457CJ56	LLY	0.094	0.364	LLM
US532457CJ56	LLY	-0.295	0.149	NLP
US532457CK20	LLY	0.087	0.250	LLM
US532457CK20	LLY	-0.192	0.319	NLP
US532457CP17	LLY	0.024	0.746	LLM
US532457CP17	LLY	0.042	0.788	NLP
US532457CQ99	LLY	-0.030	0.716	LLM
US532457CQ99	LLY	0.022	0.858	NLP
US30303M8B15	META	0.012	0.521	LLM
US30303M8B15	META	0.017	0.514	NLP
US30303M8D70	META	0.014	0.430	LLM
US30303M8D70	META	0.011	0.622	NLP
US30303M8G02	META	0.023	0.227	LLM
US30303M8G02	META	0.027	0.322	NLP
US30303M8H84	META	0.016	0.385	LLM
US30303M8H84	META	0.011	0.675	NLP
US30303M8L96	META	0.011	0.587	LLM
US30303M8L96	META	0.031	0.268	NLP
US30303M8M79	META	0.011	0.579	LLM
US30303M8M79	META	0.020	0.462	NLP
US30303M8S40	META	0.037	0.521	LLM
US30303M8S40	META	0.080	0.367	NLP
US30303M8T23	META	0.049	0.373	LLM
US30303M8T23	META	0.103	0.180	NLP
USU59197AB66	META	0.016	0.397	LLM
USU59197AB66	META	0.017	0.502	NLP
USU59197AD23	META	0.015	0.392	LLM

ISIN	Ticker	Estimate	p-value	Sentiment Type
USU59197AD23	META	0.011	0.623	NLP
US00507VAK52	MSFT	-0.004	0.834	LLM
US00507VAK52	MSFT	0.003	0.903	NLP
US594918BB90	MSFT	-0.004	0.787	LLM
US594918BB90	MSFT	0.034	0.078	NLP
US594918BJ27	MSFT	-0.015	0.289	LLM
US594918BJ27	MSFT	0.014	0.455	NLP
US594918BR43	MSFT	-0.017	0.279	LLM
US594918BR43	MSFT	0.000	0.992	NLP
US594918BY93	MSFT	-0.015	0.276	LLM
US594918BY93	MSFT	0.004	0.832	NLP
US594918CG78	MSFT	0.052	0.225	LLM
US594918CG78	MSFT	-0.035	0.423	NLP
US594918CN20	MSFT	0.079	0.594	LLM
US594918CN20	MSFT	-0.028	0.721	NLP
US67020YAM21	MSFT	0.074	0.250	LLM
US67020YAM21	MSFT	0.044	0.649	NLP
US67020YAN04	MSFT	0.083	0.244	LLM
US67020YAN04	MSFT	0.080	0.442	NLP
USU59340AH90	MSFT	0.032	0.471	LLM
USU59340AH90	MSFT	-0.037	0.411	NLP
US64110LAL09	NFLX	-0.022	0.299	LLM
US64110LAL09	NFLX	0.003	0.925	NLP
US64110LAN64	NFLX	-0.017	0.464	LLM
US64110LAN64	NFLX	0.046	0.211	NLP
US64110LAS51	NFLX	-0.025	0.387	LLM
US64110LAS51	NFLX	0.062	0.131	NLP
US64110LAT35	NFLX	-0.040	0.134	LLM
US64110LAT35	NFLX	0.035	0.396	NLP
US64110LAU08	NFLX	-0.026	0.277	LLM
US64110LAU08	NFLX	0.056	0.130	NLP
US64110LAV80	NFLX	-0.037	0.153	LLM
US64110LAV80	NFLX	0.047	0.219	NLP
US64110LAX47	NFLX	-0.038	0.190	LLM
US64110LAX47	NFLX	0.039	0.358	NLP
US64110LAY20	NFLX	-0.054	0.070	LLM
US64110LAY20	NFLX	0.051	0.252	NLP
USU74079AN15	NFLX	-0.019	0.433	LLM
USU74079AN15	NFLX	0.052	0.170	NLP
USU74079AT84	NFLX	-0.068	0.025	LLM
USU74079AT84	NFLX	0.046	0.311	NLP
US67066GAE44	NVDA	0.012	0.328	LLM
US67066GAE44	NVDA	0.002	0.938	NLP
US67066GAF19	NVDA	-0.023	0.151	LLM
US67066GAF19	NVDA	-0.023	0.326	NLP
US67066GAG91	NVDA	-0.015	0.313	LLM
US67066GAG91	NVDA	-0.012	0.579	NLP

ISIN	Ticker	Estimate	p-value	Sentiment Type
US67066GAH74	NVDA	-0.019	0.181	LLM
US67066GAH74	NVDA	-0.017	0.451	NLP
US67066GAJ31	NVDA	-0.021	0.171	LLM
US67066GAJ31	NVDA	-0.015	0.493	NLP
US67066GAM69	NVDA	-0.002	0.938	LLM
US67066GAM69	NVDA	-0.003	0.923	NLP
US67066GAN43	NVDA	-0.013	0.500	LLM
US67066GAN43	NVDA	0.010	0.754	NLP
US83417KAD00	TSLA	-0.012	0.405	LLM
US83417KAD00	TSLA	0.010	0.690	NLP
US83417KAJ79	TSLA	-0.011	0.548	LLM
US83417KAJ79	TSLA	-0.010	0.761	NLP
US83417KAP30	TSLA	-0.011	0.501	LLM
US83417KAP30	TSLA	-0.014	0.633	NLP
US83417KAU25	TSLA	-0.011	0.514	LLM
US83417KAU25	TSLA	-0.016	0.603	NLP
US83417KAZ12	TSLA	-0.012	0.459	LLM
US83417KAZ12	TSLA	-0.017	0.547	NLP
US83417KBC18	TSLA	-0.015	0.369	LLM
US83417KBC18	TSLA	-0.021	0.469	NLP
US83417KBH05	TSLA	-0.007	0.617	LLM
US83417KBH05	TSLA	-0.006	0.789	NLP
US83417KBM99	TSLA	-0.007	0.606	LLM
US83417KBM99	TSLA	-0.007	0.784	NLP
US83417KBR86	TSLA	-0.040	0.487	LLM
US83417KBR86	TSLA	-0.083	0.419	NLP
US83417KBV98	TSLA	-0.007	0.596	LLM
US83417KBV98	TSLA	-0.006	0.797	NLP
US92826CAD48	V	-0.002	0.842	LLM
US92826CAD48	V	0.001	0.958	NLP
US92826CAE21	V	-0.003	0.779	LLM
US92826CAE21	V	-0.017	0.301	NLP
US92826CAF95	V	0.002	0.818	LLM
US92826CAF95	V	-0.011	0.492	NLP
US92826CAH51	V	-0.018	0.088	LLM
US92826CAH51	V	-0.031	0.103	NLP
US92826CAJ18	V	-0.004	0.713	LLM
US92826CAJ18	V	-0.025	0.180	NLP
US92826CAK80	V	-0.023	0.116	LLM
US92826CAK80	V	-0.024	0.282	NLP
US92826CAL63	V	-0.031	0.052	LLM
US92826CAL63	V	-0.022	0.363	NLP
US92826CAM47	V	-0.018	0.239	LLM
US92826CAM47	V	-0.031	0.195	NLP
US92826CAN20	V	-0.005	0.768	LLM
US92826CAN20	V	-0.037	0.141	NLP
US92826CAP77	V	-0.011	0.523	LLM

ISIN	Ticker	Estimate	p-value	Sentiment Type
US92826CAP77	V	-0.023	0.351	NLP
US931142CD32	WMT	0.002	0.683	LLM
US931142CD32	WMT	-0.014	0.108	NLP
US931142CH46	WMT	0.021	0.052	LLM
US931142CH46	WMT	0.015	0.349	NLP
US931142ED14	WMT	0.029	0.059	LLM
US931142ED14	WMT	0.014	0.517	NLP
US931142EE96	WMT	0.030	0.044	LLM
US931142EE96	WMT	0.013	0.532	NLP
US931142EM13	WMT	0.036	0.040	LLM
US931142EM13	WMT	0.018	0.470	NLP
US931142ER00	WMT	0.033	0.124	LLM
US931142ER00	WMT	0.008	0.791	NLP
US931142EW94	WMT	0.042	0.079	LLM
US931142EW94	WMT	0.007	0.816	NLP
US931142EX77	WMT	0.048	0.070	LLM
US931142EX77	WMT	0.004	0.906	NLP
US931142FA65	WMT	0.053	0.078	LLM
US931142FA65	WMT	-0.053	0.188	NLP
US931142FB49	WMT	0.031	0.277	LLM
US931142FB49	WMT	-0.040	0.332	NLP
US30231GAF90	XOM	0.006	0.495	LLM
US30231GAF90	XOM	0.003	0.860	NLP
US30231GAT94	XOM	0.009	0.400	LLM
US30231GAT94	XOM	0.019	0.257	NLP
US30231GBD34	XOM	0.018	0.193	LLM
US30231GBD34	XOM	0.012	0.580	NLP
US30231GBH48	XOM	0.026	0.036	LLM
US30231GBH48	XOM	0.013	0.554	NLP
US30231GBJ04	XOM	0.028	0.022	LLM
US30231GBJ04	XOM	0.025	0.256	NLP
US701885AJ44	XOM	0.085	0.066	LLM
US701885AJ44	XOM	0.114	0.171	NLP
US723787AB37	XOM	0.027	0.050	LLM
US723787AB37	XOM	0.031	0.194	NLP
US723787AT45	XOM	0.048	0.002	LLM
US723787AT45	XOM	0.030	0.296	NLP
US723787AV90	XOM	0.027	0.109	LLM
US723787AV90	XOM	0.001	0.973	NLP
USU7024PAH11	XOM	0.032	0.298	LLM
USU7024PAH11	XOM	0.061	0.344	NLP

D Appendix D

Ticker	Sentiment	Serial	ARCH	Normality			Stability
				JB	Skewness	Kurtosis	
AAPL	LLM	0.900	0.400	0.000	0.200	0.000	0.300
	NLP	1.000	0.400	0.000	0.000	0.000	0.500
AMZN	LLM	0.900	0.500	0.000	0.200	0.000	0.600
	NLP	1.000	0.500	0.000	0.000	0.000	0.600
AVGO	LLM	0.900	0.600	0.000	0.100	0.000	0.900
	NLP	0.800	0.800	0.000	0.100	0.000	0.800
BRK	LLM	1.000	0.375	0.000	0.000	0.000	0.750
	NLP	1.000	0.375	0.000	0.000	0.000	0.875
GOOGL	LLM	0.800	0.600	0.000	0.000	0.000	0.700
	NLP	0.800	0.400	0.000	0.000	0.000	0.700
LLY	LLM	0.700	0.300	0.400	0.400	0.400	0.600
	NLP	0.500	0.300	0.300	0.400	0.300	0.700
META	LLM	1.000	0.800	0.100	0.800	0.200	0.500
	NLP	1.000	0.800	0.100	0.500	0.100	0.500
MSFT	LLM	0.700	0.300	0.100	0.300	0.100	0.500
	NLP	0.200	0.300	0.100	0.100	0.100	0.600
NFLX	LLM	0.400	0.100	0.000	0.000	0.000	0.100
	NLP	1.000	0.200	0.000	0.000	0.000	0.200
NVDA	LLM	1.000	0.000	0.000	0.286	0.000	0.286
	NLP	1.000	0.714	0.000	0.000	0.000	0.286
TSLA	LLM	1.000	0.000	0.000	0.000	0.000	0.700
	NLP	1.000	0.000	0.000	0.000	0.000	0.600
V	LLM	0.600	0.300	0.000	0.000	0.000	0.400
	NLP	1.000	0.400	0.000	0.300	0.000	0.400
WMT	LLM	1.000	0.600	0.000	0.000	0.000	0.800
	NLP	0.800	0.500	0.000	0.000	0.000	0.200
XOM	LLM	0.900	0.300	0.000	0.000	0.000	0.300
	NLP	0.800	0.500	0.000	0.100	0.000	0.300

Table D.1: VAR Diagnostic Test Results

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